Assignment (19.3)

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Task 1 – Python to C++ Conversion

Prompt:
Convert a Python Student class into C++.

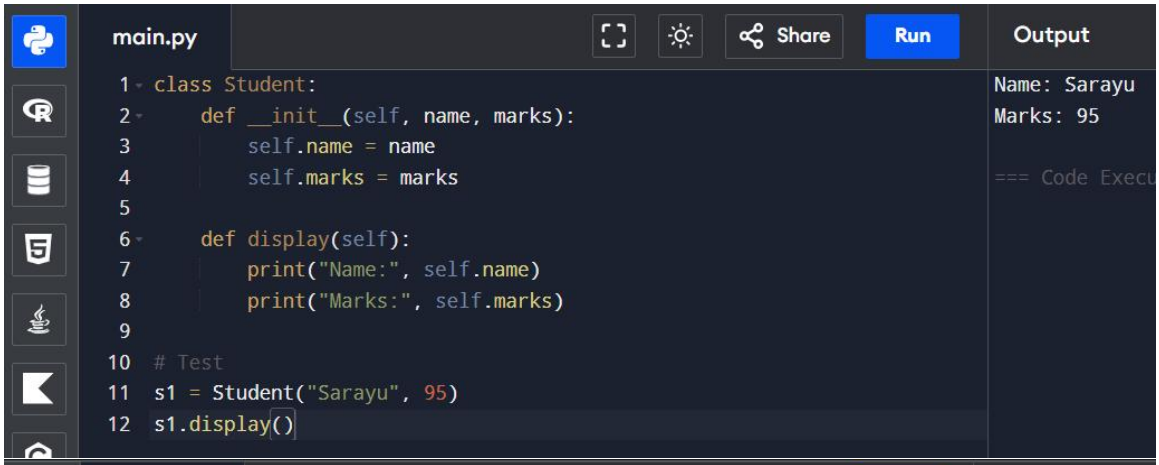
Python Code:

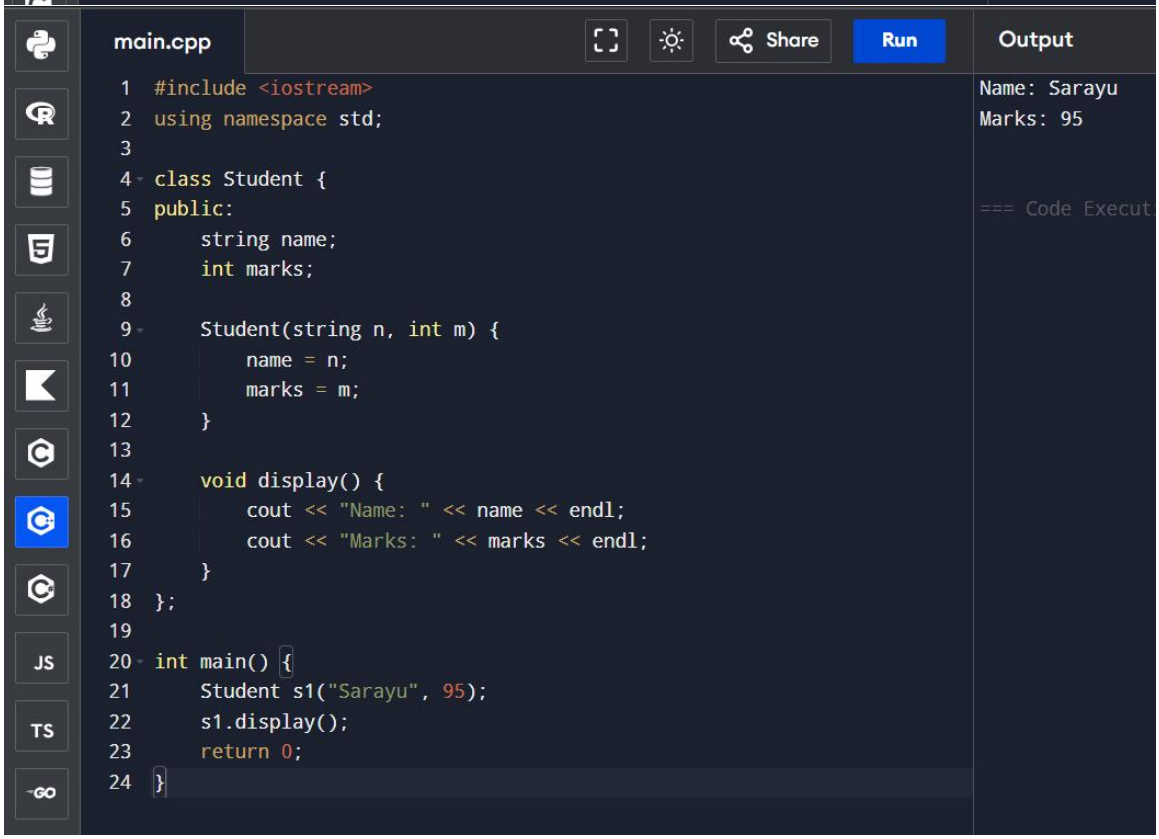
```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks
    def display(self):
        print(self.name, self.marks)
```

C++ Code:

```
#include <iostream>
using namespace std;
class Student {
public:
    string name;
    int marks;
    Student(string n, int m){
        name = n; marks = m;
    }
    void display(){
        cout << name << " " << marks;
    }
};
```

Output:
Same output in both languages





Task 2 – Java to Python Conversion

Prompt:
Convert Java isPrime() method into Python.

Java Code:

```
boolean isPrime(int n){
  if(n<=1) return false;
```

```
    for(int i=2;i<n;i++){
        if(n%i==0) return false;
    }
    return true;
}
```

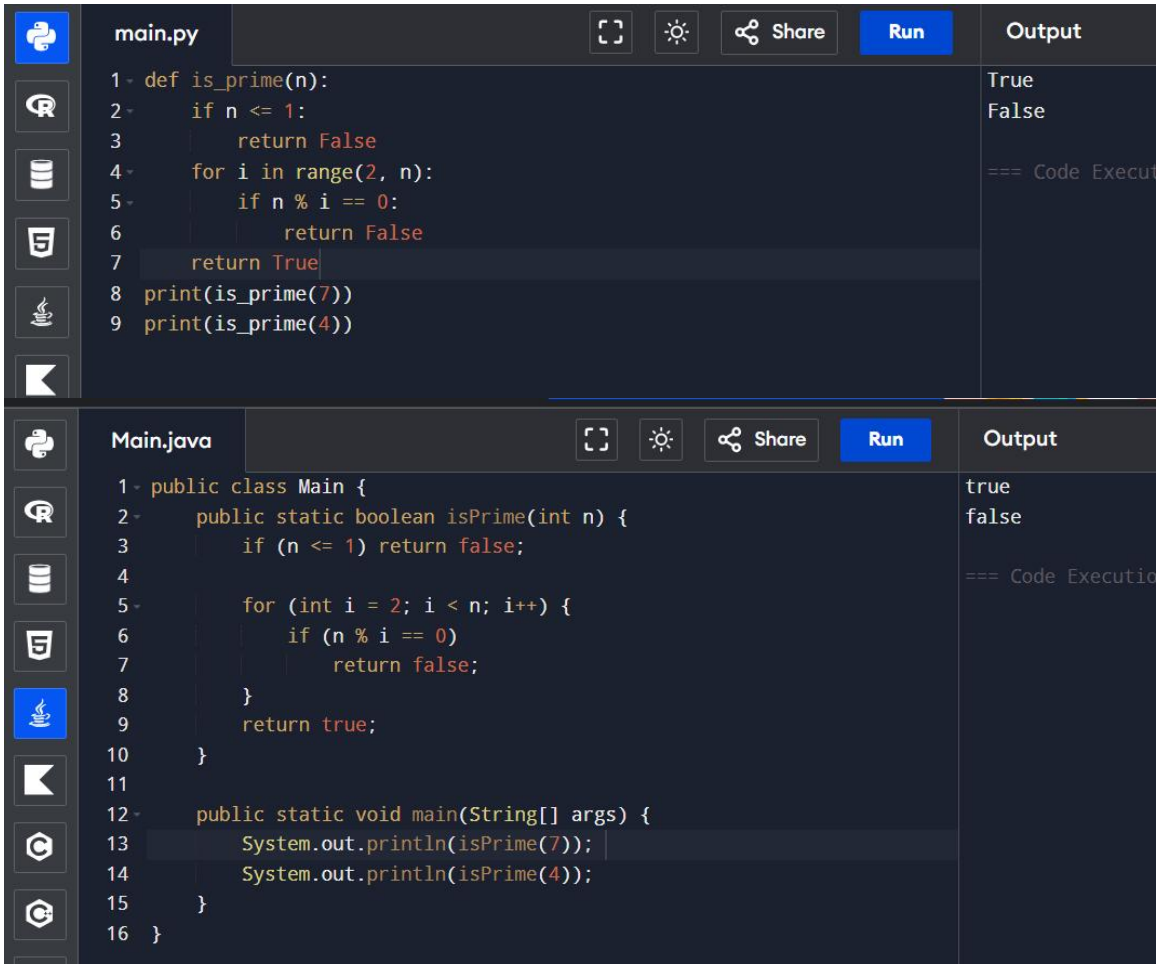
Python Code:

```
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True
```

Output:

Input: 7 → True

Input: 4 → False



Task 3 – Pseudocode to Python

Prompt:
Convert bubble sort pseudocode to Python.

Python Code:

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(0, n-i-1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
    return arr

print(bubble_sort([5,2,9,1]))
```

Output:
[1, 2, 5, 9]

Task 4 – SQL to Pandas

Prompt:
Convert SQL query to Pandas.

SQL:
SELECT name, salary FROM employees WHERE salary > 50000

Python Code:

```
import pandas as pd
data = {"name":["A","B","C"],"salary":[40000,60000,70000]}
df = pd.DataFrame(data)
result = df[df["salary"] > 50000][["name","salary"]]
print(result)
```

Output:
B 60000
C 70000

Task 5 – Algorithm Translation Across Languages

Prompt:
Convert Python linear search to C.

Python Code:

```
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1
```

C Code:

```
#include <stdio.h>
int linearSearch(int arr[], int n, int key){
    for(int i=0;i<n;i++){
        if(arr[i]==key) return i;
    }
    return -1;
}
```

Challenges:
- Syntax differences
- Memory management
- Library support

main.py

Share Run Outp

```python
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1

# Test
print(linear_search([10, 20, 30, 40], 30))
```

```
2

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```

main.c

Share Run Output

```c
#include <stdio.h>

int linearSearch(int arr[], int n, int key) {
    for(int i = 0; i < n; i++) {
        if(arr[i] == key)
            return i;
    }
    return -1;
}

int main() {
    int arr[] = {10, 20, 30, 40};
    int result = linearSearch(arr, 4, 30);

    printf("Index: %d", result);
    return 0;
}
```

```
Index: 2

=== Code Exe
```